

AI Engineer Assessment Submission

Local document intelligence, structured computation and MCP tool calling

Submission detail	Value
Candidate	Akshita Rastogi
Role	AI Engineer
Runtime	Local Docker Compose and Ollama
Interface	Streamlit with FastAPI
Data	3 official PDFs, 603 pages, 70,402 OpenFEMA rows
Core stores	MongoDB and Qdrant

The solution meets every required capability and adds measured retrieval, bounded memory, tenant isolation, auditability and observability without hiding the core logic.

1 Problem and solution

The assessment requires a lightweight assistant that answers document questions, calculates answers from structured data, calls an external tool, handles uncertainty, collects feedback and reports evaluation results. SupplyGuard AI implements these as explicit execution paths.

The use case combines procurement, supply-chain cybersecurity and disaster risk. It creates one coherent scenario while exercising policies, tables, acronyms, geographic data and repeated records.

Solution summary

Layer	Implementation	Responsibility
Experience	Streamlit	Multi-turn questions, citations, traces and per-answer feedback
API	FastAPI	Contracts, auth, status codes, orchestration and health
Models	Ollama	Local intent routing, generation and 768-dimensional embeddings
Operational data	MongoDB	Chunks, FEMA rows, memory, feedback, audits and runs
Vector search	Qdrant	Persistent cosine vectors and tenant-filtered payloads
External tool	MCP server	Validated FEMA queries over Streamable HTTP

Required capability mapping

Requirement	Delivered behaviour	Reviewer evidence
Document assistant	Hybrid RAG with evidence-only generation	Answer plus page-level citations
Structured data	Allowlisted MongoDB computation	Computation trace and typed result
External tool	Independent MCP service	Validated arguments and exact output
Reliability	Scope guard and retrieval refusal	Zero-confidence unsupported response
Feedback	Idempotent MongoDB record	Helpful or Not Helpful per request
Evaluation	8 required plus 6 difficult cases	Expected, actual and pass or fail

2 Architecture and end-to-end flow

The editable architecture is supplied as docs/SupplyGuard_AI_Architecture.drawio. It separates offline ingestion from online questions and shows all four routing outcomes.

Offline ingestion

1. Read only the official PDFs and OpenFEMA CSV under the data directory.
2. Extract page text in reading order, serialise tables separately and count images.
3. Mark blank and OCR-required pages rather than silently discarding them.
4. Create page-element-bounded chunks and generate embeddings locally through Ollama.
5. Upsert chunks into MongoDB and vectors into Qdrant using deterministic IDs.
6. Store typed FEMA rows, checksums, ingestion totals and parser failures.

Online question flow

1. Streamlit sends the tenant, conversation ID and question to FastAPI.
2. The API validates the request, loads bounded referential memory and assigns a request ID.
3. A local LLM proposes document, analytics, MCP or unsupported; rules authorise the route.
4. The selected path retrieves evidence, computes, calls MCP or refuses.
5. The orchestrator returns answer, confidence, citations, warnings and execution trace.
6. MongoDB stores the audit and conversation turn; feedback is stored by request ID.

Why explicit orchestration

Three execution routes plus refusal do not require a graph framework. A typed orchestrator is easier to inspect, debug and test. LangGraph would become valuable for resumable workflows, parallel tools or human approval checkpoints.

Local model selection

Mistral instruct was selected from the locally installed models for evidence-only instruction following and refusal. It runs at temperature zero. nomic-embed-text is purpose-built for retrieval and produces 768-dimensional vectors. Ollama keeps data local and avoids paid APIs.

3 Dataset, chunking and retrieval

Official mixed-content corpus

Source	Size	Why it is difficult
World Bank Procurement Regulations 2025	154 pages	Rules, annexes and specialised terminology
World Bank Contract Management Guidance 2024	124 pages	Tables, diagrams, KPIs and workflows
NIST SP 800-161 Rev. 1 Update 1	325 pages	Dense controls, acronyms and cross-references
OpenFEMA Disaster Declarations v2	70,402 rows	Nulls, dates, geography and repeated rows

Measured chunking decision

The implementation uses 160 words with 30 words of overlap, a 130-word step. Chunks stay inside each page element, and tables remain separate from prose. The value was selected from a reproducible benchmark rather than convention.

Candidate	Chunks	Recall at 6	Hit rate	MRR at 6	Context words
160 / 30 selected	2,198	74.6%	100%	0.413	749
240 / 40	1,562	70.5%	100%	0.396	1,067
400 / 60	1,114	74.1%	100%	0.479	1,434

The selected candidate achieved the highest concept recall with the smallest evidence packet. The 400-word candidate had better MRR but consumed 91 percent more context. Heading hierarchy is inconsistent across publishers, so fully heading-driven splitting would be unreliable. Element-aware boundaries preserve tables and pages while fixed overlap stays measurable.

Hybrid retrieval and grounding

- Qdrant retrieves 20 dense cosine candidates for semantic paraphrases.
- MongoDB retrieves 20 lexical candidates for acronyms, controls and exact values.
- Reciprocal Rank Fusion combines ranks without mixing incompatible score scales.
- The top six chunks form the evidence packet; evidence below 0.35 is refused.
- Citations come from retrieved records, not model-generated source text.

4 Structured analytics and MCP

Correct structured computation

The LLM never performs FEMA arithmetic. The service maps supported wording to known fields and builds an allowlisted MongoDB pipeline. User text cannot supply operators or arbitrary pipelines.

OpenFEMA stores one row per designated area, so one disaster appears multiple times. Raw-row counting incorrectly reports Texas with 5,444. Grouping by state and disasterNumber first correctly returns California with 397 distinct declarations. This is a regression case.

External tool through MCP

The MCP server is a separate process using Streamable HTTP, providing a real client-server boundary instead of labelling a local function as a tool.

Tool	Validated inputs	Output
disaster_statistics	Optional state and year	Distinct totals, row count, incident breakdown and source
lookup_disasters	State, year and bounded limit	Matching records with explicit source

Guarded intent selection

Explicit FEMA tool wording selects MCP. Otherwise, a local LLM proposes an intent and rules control access. FEMA analytics requires a FEMA subject and supported operation. Procurement misrouted as analytics is corrected to document RAG. Unrelated requests are refused before access.

Failure handling

- Invalid state or year values fail validation before tool execution.
- Tool and storage failures produce controlled warnings and dependency status.
- No failed path falls back to unsupported model knowledge.
- The response exposes the tool, validated arguments, source and output.

5 Reliability, memory and feedback

Reliability controls

Risk or edge case	Control
Weak evidence	Refuse below threshold; never generate from model memory
Prompt injection	Treat chunks as untrusted evidence, not application instructions
NoSQL injection	Allowlisted fields and server-owned pipelines
Cross-tenant leakage	Tenant filters in MongoDB and Qdrant plus isolated conversation IDs
Duplicate ingestion	Deterministic IDs and idempotent upserts
Duplicate feedback	Request-ID uniqueness and disabled submitted state
Conflicting evidence	Numeric contradiction warning
Scanned or blank pages	Explicit OCR-required or blank classification
Misclassified intent	Deterministic eligibility corrects or refuses the proposal

Context and memory management

Only referential follow-ups reuse history. Memory is isolated by tenant and conversation, bounded to six turns and 4,000 characters, and expires after 24 hours through MongoDB TTL. It resolves 'What about 2024?' after a California 2025 question, but never replaces evidence.

Feedback and learning

Every answer has independent Helpful or Not Helpful feedback with an optional reason. The UI confirms success and disables duplicates. This is not reinforcement learning or RLHF because no weights, rewards, policy or prompts change online. It is labelled data for reviewed regression tests.

API and observability

- Typed request and response models with validation, authentication and conflict status codes.
- Structured JSON logs, correlation IDs, duration and Prometheus metrics.
- Dependency-aware readiness for MongoDB, Qdrant, Ollama and MCP.
- Replayable audits containing routing, evidence, computation and tool decisions.

6 Evaluation and additional engineering

Evaluation design

The suite contains four document questions, two structured questions, one MCP question and one refusal case as required, plus six difficult regressions. It records Expected, Actual and Pass or Fail. Document cases measure expected-concept completeness, excerpt-supported groundedness and citation coverage. Other routes measure computation, tool behaviour and refusal correctness. The completeness and groundedness threshold is 0.50.

The chunk benchmark is an offline lexical proxy; the end-to-end suite exercises routing, retrieval, generation, computation, tool use and refusal. The separation avoids presenting retrieval metrics as full answer accuracy.

Features beyond the assessment

Extension	Engineering value
Hybrid RRF retrieval	Combines semantic recall and exact-term precision using inspectable ranks
Measured chunking	Uses recall, hit rate, MRR, latency and context cost
Bounded memory	Natural follow-ups without unbounded or cross-tenant context
Independent MCP	Actual client-server boundary and typed validation
Correct FEMA grain	Prevents a confident wrong answer from repeated rows
Replayable audits	Preserves route, evidence and tool decisions
Security controls	Prompt boundary, allowlisted analytics and scope guards
Operational visibility	Logs, request IDs, metrics, health and ingestion records
Reproducibility	Official snapshots, checksums and deterministic IDs

7 Limitations and review guide

Known limitations

- Scanned pages are detected but not OCR processed.
- The lexical retriever is lightweight and should become BM25 for a larger corpus.
- The retrieval threshold is practical but not statistically calibrated.
- API keys and container settings are demonstration credentials.
- Public datasets are reproducible snapshots rather than continuously refreshed feeds.

Next improvements

- Add OCR with layout-aware table and figure extraction.
- Add BM25 and a local cross-encoder reranker.
- Create labelled relevance judgements and calibrate confidence.
- Expand integration, concurrency and dependency-failure tests.
- Add scheduled source refresh with checksum and schema-drift validation.
- Replace demo credentials with OIDC and production secret management.

Recommended reviewer flow

1. Open the Draw.io architecture and review offline versus online responsibilities.
2. Verify Docker readiness and persistent storage counts.
3. Ask a cited World Bank question and inspect the hybrid trace.
4. Ask which state has the most declarations and confirm California 397.
5. Call the FEMA MCP tool for California in 2025, then ask about 2024.
6. Ask an unrelated identity question and verify refusal.
7. Submit feedback, inspect MongoDB and view a vector payload in Qdrant.
8. Open Evaluation and compare expected, actual and pass or fail.

Submission artifacts

Artifact	Purpose
README.md	Setup, design decisions, commands and limitations
docs/SupplyGuard_AI_Architecture.drawio	Editable end-to-end diagram
SupplyGuard_AI_Assessment_Submission.docx	Reviewer-oriented submission narrative
docs/DEMO_SPEECH.md	Timed presentation narration
docs/DATA_MANIFEST.md	Official source provenance and checksums